



PORTFOLIO

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ABOUT ME



I'm a mechanical engineer trained at TU Delft and UC Berkeley who builds hardware that works in the real world, with projects ranging from a low-cost autorefractor to a CNC machine I designed and built end to end.

I work best at the intersection of mechanical design and validation, taking system from analysis to functional hardware under tight timelines.

Belgian-born and fluent in five languages, I'm comfortable in fast-paced, multidisciplinary teams across different engineering cultures. I'm looking to do that kind of work on hard problems in medical devices, energy, or other technically demanding fields.

When I'm not designing or debugging something, you'll find me outdoors!

RESUME

Portfolio | LinkedIn | Mail: _michael_rbn@berkeley.edu | Phone: +1 (929) 512-0901

Driven MechE graduate with hands-on experience prototyping, testing, and iterating real-world mechanical systems. Motivated to apply my technical depth and creative problem-solving in a **design or manufacturing role**. With projects taken from analysis to functional hardware under tight timelines, I bring a strong background in **engineering analysis, rapid prototyping and design optimization**. Fluent in **5 languages**, I collaborate effectively across multidisciplinary teams in fast-paced environments.

Mechanical Engineer | MPC Lab/C Berkeley – Berkeley, CA, USA June – Present

- Contract role supporting NIWC Pacific on development of an **all-terrain autonomous surveillance rover** (Project LUCI).
- Redesigning 3D-printed chassis of the rover to **improve structural robustness and usability**.
- Developed and implemented **testing protocols** to validate the control stack for autonomous navigation and localization.

Chassis Engineer - Driver Safety Lead | CALSOL Solar Vehicle Team – Berkeley, CA, USA Sept 2025 – Present

- Responsible for GenXII seatbelt implementation in the occupant cell, ensuring **ASC and WSC regulation compliance**.
- Simplified shoulder belt anchorage, enabling in-house manufacturing. **Reduced weight by 40%** using topology optimization.
- Designed and executed **Instron validation tests** ensuring mounts withstand crash load with 1.6 FOS.
- Managed large CAD assemblies, prepared drawing packages using **GD&T** and collaborated with suppliers.

AutoCAD Drawing Engineer | KBC Acuity – Remote Oct 2022 – June 2025

- Produced over **300 technical drawings** for nationwide security camera installations at KBC Bank offices.

Teaching Assistant (UGS: Statics, Mechanics of Materials) | TU Delft – Delft, ZH, The Netherlands Sept 2022 – Jan 2024

Handheld Autoforetraction Device UC Berkeley - Continued Independently • Led optical design of a Scheiner disk based vision screening system to ± 0.500 SPH accuracy. Secured \$1K grant for prototyping. • Created alignment jigs using SolidWorks and developed calibration protocols to verify and maintain system accuracy. • Supported development of an image-processing pipeline to compute corrective lens power.	Jan 2025 – Present
Tactile Robotic End Effector, Graduate Capstone Embodied Dexterity Group – UC Berkeley • Scaled a multi-chamber suction based system from a research-grade prototype to a production-ready unit for 1,000+ orders. • Standardized the robot-arm interface, reduced part count 32 $\rightarrow$ 9 and cut setup time 15min $\rightarrow$ 1:45min. • Applied design-for-manufacturability principles and FMEA to redesign injection-molded parts for better serviceability and reliability. • Designed and integrated custom PCB with BOM management using KiCad while ensuring clean cable routing.	Sept 2025 – May 2026
Light module for Robotic Surgery System, BS Thesis AdLap Lab – TU Delft • Designed operating room compliant detachable light module for a low-cost laparoscopic system, with white/IR mode switching. • Achieved 3x illuminance while reducing device size by 50% relative to prior prototype. • Implemented mechanically actuated mode switching with precise alignment, guided by 2D ray tracing and optical bench tests. • Engineered thermal management system using heat transfer analysis and validating performance via long-duration thermal testing.	Jan 2024 – June 2024
CNC machine TU Delft • Designed and built a CNC machine for PMMA carving using an H-bot single-belt gantry architecture; • Cut footprint by 10% vs benchmark 3-axis system and minimized part count. Placed top 5% in faculty competition. • Integrated belt pre-tensioning and vibration damping to improve motion control and reliability.	Sept 2023 – Dec 2023

Mechanical: CAD & Drawing (SolidWorks CSWA certified), FMEA, Tolerancing, GD&T (ASME Y14.5-2018), DFx, RCA, V&V, Material Selection (Granta), BOM Management, Motor & Sensor Selection, Thermal Optimization, Controls (PID, MPC), Assembly Modeling
Manufacturing: CNC, Additive (FFF, Cura), Injection Molding, Sheet Metal Fabrication, Bench Tests, Machine Shop Trained
Software: FEA (Abaqus, Ansys, COMSOL), Python, MATLAB, C++, G-Code, Git, Ros2, VSCode, Linux, Excel (PivotTables)
Languages & Soft Skills: English (near-native, TOEFL 114/120), French (native), Dutch (native), Spanish (fluent), Hebrew (basic), Reporting & Documentation, Proactive, Adaptability, Collaboration, Task Prioritization, Creativity

University of California at Berkeley (UC Berkeley) – GPA 3.8/4	Berkeley, CA, USA
Master's degree in Mechanical Engineering – Biomechanics & Control	Aug 2025 – May 2026
Delft University of Technology (TU Delft) – GPA 3.8/4	Delft, The Netherlands
Honours Bachelor's degree in Mechanical Engineering	Aug 2021 – June 2024
Exchange semester at Norwegian University of Science and Technology (NTNU) – GPA 3.825/4	

Graduate Scholarship | Belgian American Education Foundation Aug 2025 – May 2026
One of 96 Belgians awarded a national grant to pursue graduate studies in the United States; covering full tuition and stipend.

Honours Program Delft Certificate | TU Delft Sept 2022 – June 2024
Selected as 1 of 4 MechE students for exclusive program; completed 20 additional credits within nominal graduation time.

https://drive.google.com/file/d/1Cl_wKEaFP7BnYoLp83KJAmWXBudB7E-B/view?usp=sharing

EDUCATION



MEng Mechanical Engineer: Biomechanics & Control

Class of 2026

- Awarded scholarship covering full tuition + stipend by the Belgian American Education Foundation
- UC Berkeley Solar Vehicle Team
- Balanced leadership and entrepreneurship courses with advanced technical electives, expertise that was directly applied in a year-long Capstone project

Courses taken

TECHNICAL

Advanced Design for the Human Body, Product Development, Robotic Locomotion, Experiential Advanced Control (MPC), Dynamics & Control of Unmanned Aerial Vehicles

BUSINESS AND LEADERSHIP

Deep Tech Commercialization, Impact Startup Disco, R&D Technology & Ethics Management, Technology Strategy for Engineering Leaders, Communication for Engineering Leaders, Product Management for Engineering Leaders



BSc Mechanical Engineer

Class of 2024

- Selected for Honours Program; completed 20 additional credits in nominal graduation time
- Drop Delft Boardsports Association
- Gained a strong design background, through several hands on projects and a broad technical program
- Exchange semester at Norwegian University of Science and Technology (NTNU) -- GPA 3.825/4

Courses taken

ENGINEERING

Solid, Continuum & Structural Mechanics, Rigid Body Dynamics, Process Engineering, Materials, Heat Transfer, Fluid Mechanics, Integrated Mechanical Systems, Control Systems Engineering, Signal Analysis

MATHEMATICS

Linear Algebra 1+2, Calculus 1+2+3, Differential Equations, Probability & Statistics, Numerical Methods & Analysis



Koninklijk Lyceum Antwerpen

Class of 2021

- Major in Latin and Mathematics
- Basketball @Mercurius BB, Jiu Jitsu @ Ikiji Ryu Dojo
- Delegated class clown



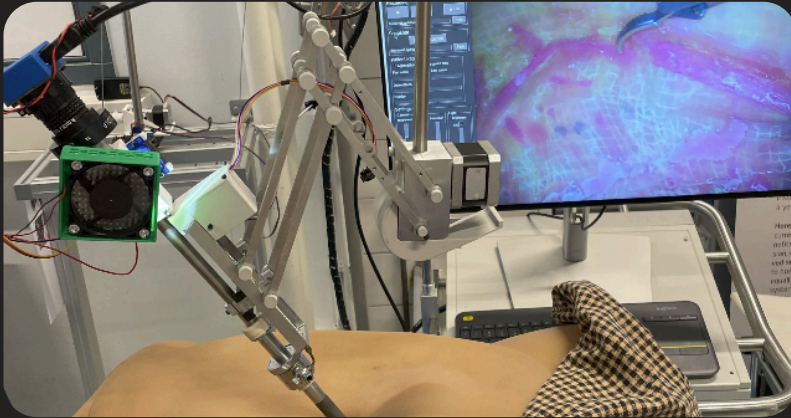
Academie Berchem

2008-2021

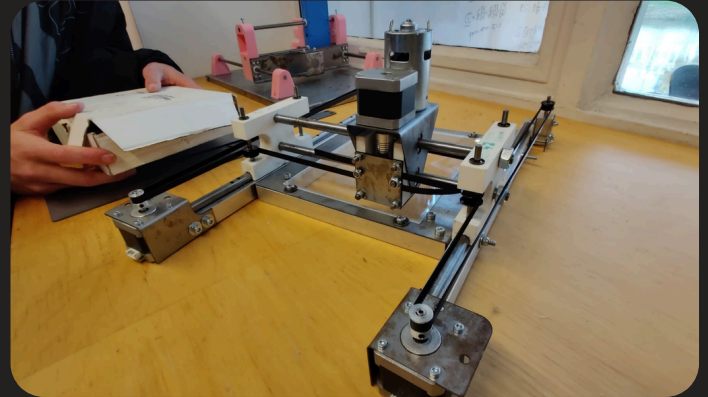
- Part time graphic arts program graduate
- Hands down my most valuable hobby growing up!
- Broadened my perspective and developed my creativity

PROJECT PORTFOLIO

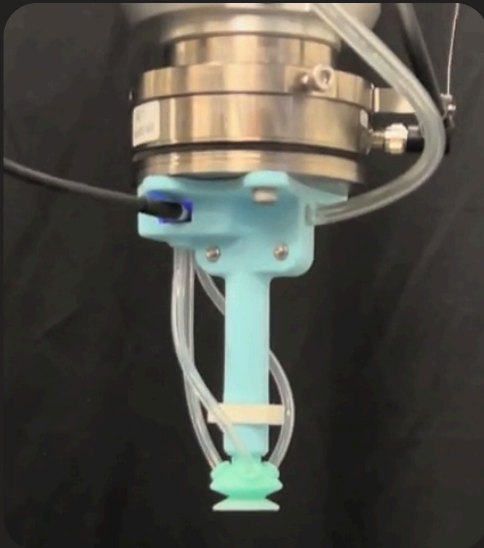
Page 7: Light Module for a Robotic Surgical System



Page 12: CNC Mill



Page 9: Tactile Robotic End Effector



I believe you learn most through hands on projects.

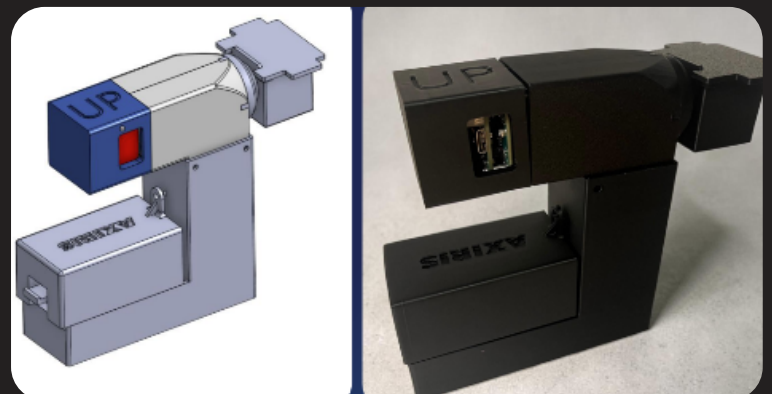
I'm not the best builder by nature, as I used to be held back by a fear of mistakes and perfectionism. That means I don't settle until I am confident in my solution working reliably, but it has held me back from starting before.

I managed to truly break this pattern when I got to university, by seeking opportunities to make my projects have an impact outside the classroom. Creating tangible solutions with people who challenge and push me to grow is what motivates me most and I'm excited to carry that mindset into my career!

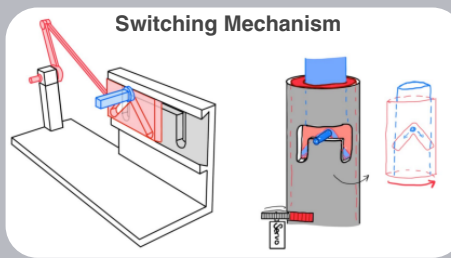
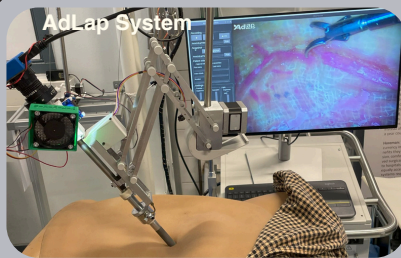
Page 10-11: CALSOL Seatbelt Design



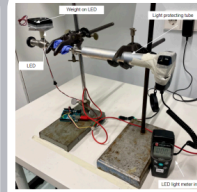
Page 8: Axiris, Handheld Autorefractor



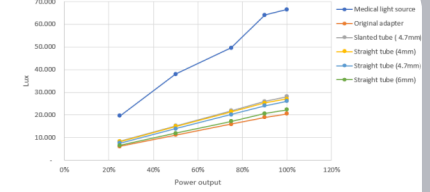
PROJECT PORTFOLIO



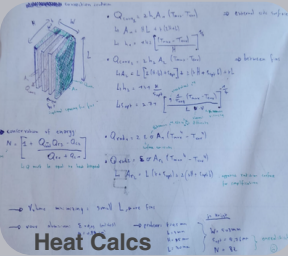
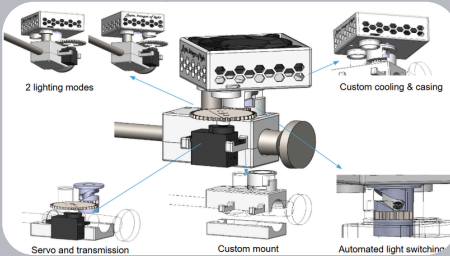
Connector Optical Test



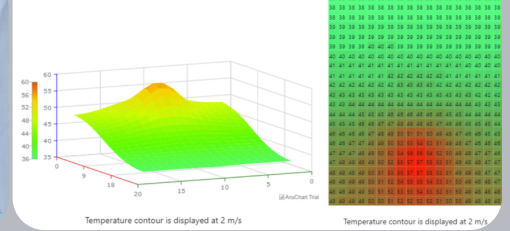
(a) Setup



(b) Results



Thermal Simulation



LIGHT MODULE FOR A ROBOTIC SURGICAL SYSTEM

Overview

For my undergraduate capstone I designed a **compact, detachable** light module for the AdLap system: a **low-cost**, fully modular laparoscopic surgery platform for low-income hospitals built around reusable components. I used **ray-tracing simulation**, **Python thermal modeling** and **bench testing** to iterate through designs.

- Target: $\geq 30\%$ of xenon reference output, $< 241 \text{ cm}^3$ volume, $< 140 \text{ g}$ weight, $\leq 41^\circ\text{C}$ surface temperature (IEC 60601-1)
- Added IR capability (780–1000 nm) for fluorescence-guided surgery, not present in the original module
- Result: 42% of benchmark reference output, within all size, weight, and thermal limits

Design Architecture

- Dual light sources on a rotating backplate: Cree XHP70.3 white LED and OSILUX 810 nm IR LED
- **Single-servo rotational switching mechanism** with precise alignment, swapping sources in $\sim 0.8 \text{ s}$ via 90° rotation
- **Polished aluminum connector tube** (28° slanted walls) couples LED output into the endoscope fiber with minimal loss
- **Milled aluminum backplate as heat spreader**, direct thermal path to an active heatsink-fan unit behind the sterile line

Design Choices

- Dropped complex lens-and-mirror optical path after first concept failed (poor beam convergence, too much light loss, high cost)
- Moved to direct LED-to-fiber alignment for simplicity, cost reduction and form factor
- Chose rotation over rail-based translation for source switching: fewer parts, more robust
- Modeled connector tube geometry in 2D ray-tracing simulation, then bench-tested the strongest candidates
- Managed a 20 W LED heat load in $< 241 \text{ cm}^3$, modeled using a Python script sweeping heatsink size, material, and airflow
 - Chose active cooling after thermal analysis showed passive fins couldn't meet the 41°C limit
 - Balanced fin surface area needed to shed heat against the airflow resistance it creates
 - Placed the fan behind the sterile line, keeping the module cleanable (ISO 13485 / ISO 8600)

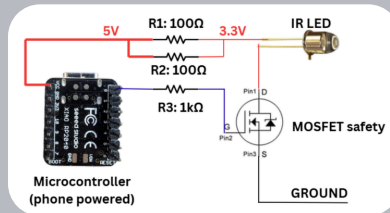
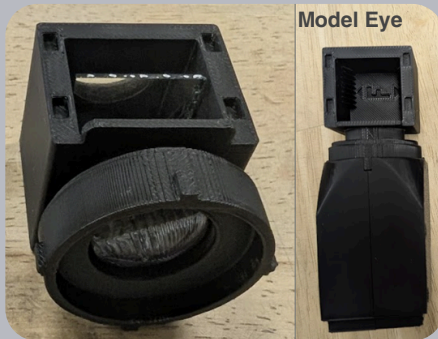
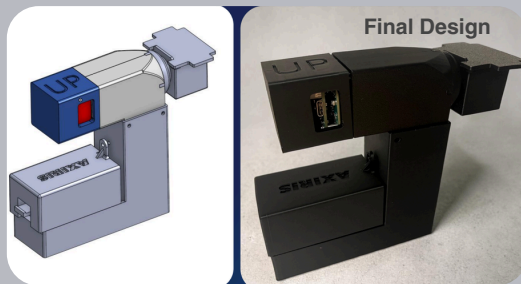
Performance & Validation

- Measured illuminance against the Karl Storz Xenon 300 clinical reference at a standardized test distance
- Bench-tested connector geometries for light coupling throughput
- Ran thermal tests confirming compliance with the $\leq 41^\circ\text{C}$ surface temperature limit

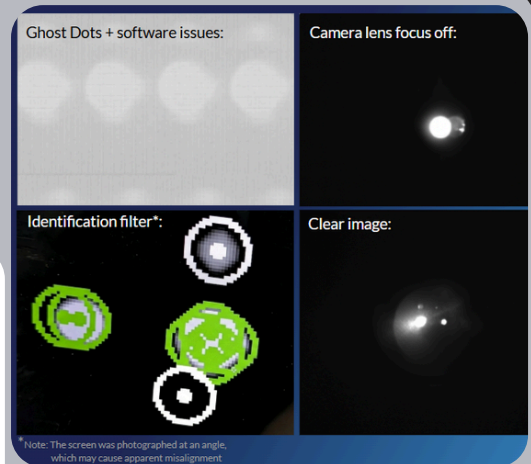
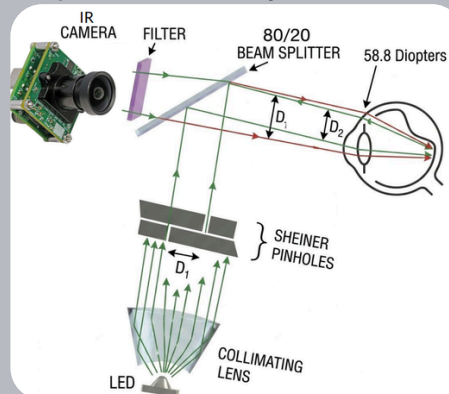
Limitations & Future Improvements

- Mount is form-fit to specific endoscope adapters; other instruments would need remodeling
- Optical collimating lens was dropped after first concept for cost and simplicity; adding one back could reduce light loss
- Laser diode sources were out of budget but would offer higher output and tuneability

PROJECT PORTFOLIO



Optical & Electrical Layout



AXIRIS: HANDHELD AUTOREFRACTOR

Overview

Served as optical lead on a handheld autorefractor built around a **dual pinhole (Scheiner-disk) approach**, taking the design through 6 prototypes over 13 weeks.

- Target: work without a clinic, grid power or an optometrist present; at \$150–\$250 retail, for use by community health workers
- Main constraint: sufficient signal in a tight enclosure while staying below ICNIRP eye-safety exposure limits
- Result: ± 0.50 Diopter accuracy, 124 x 140 x 44 mm Volume, <500 g, \$574 BOM (vs. \$5k–\$30k clinic systems)

Design Architecture & Choices

- Scheiner-disk dual-pinhole approach over standard wavefront sensor: simple geometry and image processing, low cost
- The difference in beam distance entering and exiting the eye is used to back-calculate the eye's geometry via simple optics relations
- 850nm wavelength LED to minimize retinal absorption of the signal while staying out of visible light spectrum
- 80/20 beamsplitter to prioritize signal coming out of the eye
- **Software pipeline:** contrast thresholding filter, difference of gaussian for edge detection, gaussian curve fitting for blob detection & subpixel localization, pixel-to-diopter mapping, median over multiple frames + IQR rejection with confidence score
- The system is designed to be **powered via the phone's** charging port, with data simultaneously transferred to the phone
- Switched from phone camera to a dedicated NIR camera (See3CAM_CU512M) after finding phone's IR-cut filter blocked signal
- Ground-truth calibration using **model eye** with slide-in mirror at known diopters, safe way to iterate before human testing.

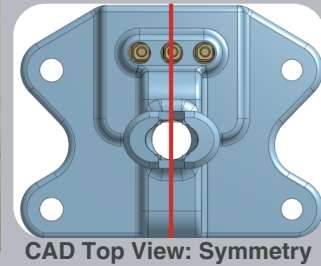
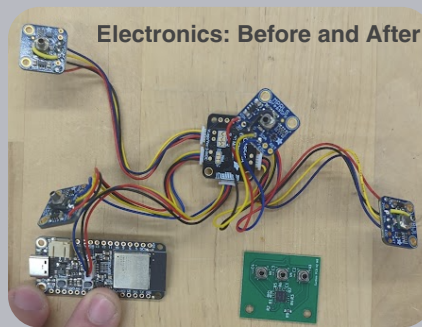
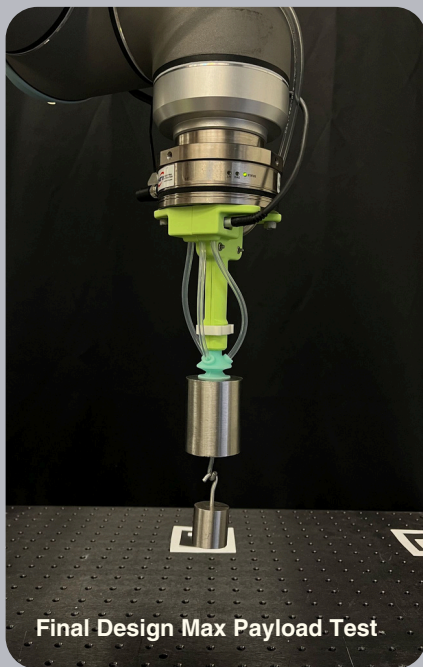
Validation & Performance

- A lot of outreach! Pugh chart to select across 50+ concepts, incorporating feedback from optometrists, users & business stakeholders
- **Bench tests** until repeatable results occurred
 - Fixed the optical hardware first, then tuned software to drive SNR up
 - Leveraged model eye's stability and modularity for root-cause debugging

Limitations & Future Improvements

- Measures 'sphere' (near/farsightedness) only; next: capture astigmatism and higher-order aberrations without losing affordability
- No durability testing; accelerated life testing is a clear next step
- Limited current from the phone using USB-C OTG cable prevents powering both the camera and LED reliably, remains unsolved
- The simple model eye was ideal for early, rough validation, but a more realistic model eye is needed for precise calibration
- This design is a proof of concept and hasn't been through a DFM pass for volume production. Step one is making a custom PCB

PROJECT PORTFOLIO



TACTILE ROBOTIC END EFFECTOR

Overview

For my graduate capstone at Berkeley's Embodied Dexterity Lab, I redesigned a research-grade tactile suction cup gripper into a **production-ready, open-source tool**, while preserving the original 2.5x grasp success improvement over vision-only grasping.

- Target: $\leq 80 \times 80 \times 60$ mm envelope, < 1 kg, ISO 9409-1 mounting, toolless cup swaps, ≥ 5000 -pick lifetime, cost $\leq \$150$
- Result: 0.72 kg max payload, 30 s cup swap, 2 min assembly time, \$45.64 BOM, reduced part count

Design Architecture

- Three-chamber silicone suction cup, each chamber feeding its own digital pressure sensor for per-chamber flow signals
- Two-layer integrated PCB consolidating the RP2040 controller, an I2C multiplexer and three pressure sensors
- Two-part PLA enclosure with a built-in compliant clip for tool-free cup access
- Standardized ISO 9409-1 robot interface: swappable for five flange variants

Design Choices

- Decided on a three-chamber layout as the best trade-off between tactile richness and manufacturability early on
- Replaced hand-wired breakout boards with a **custom PCB**, eliminating cable-related failures and shrinking form factor
- **DFA**: minimized part count, simplified handling by removing small fasteners, symmetrical design and supportless printing
- **DFM**: Split the cup mold into a five-part wedged design so one chamber releases first, fixing tearing during demolding

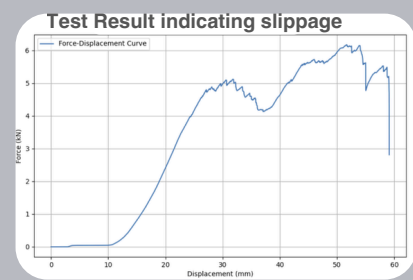
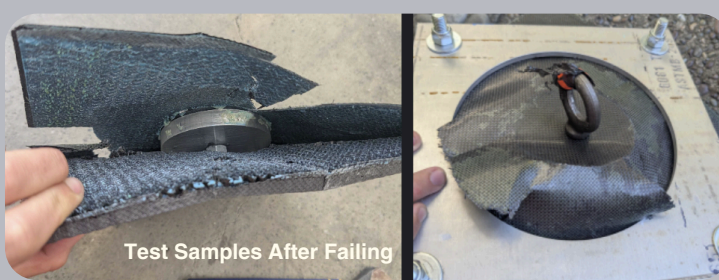
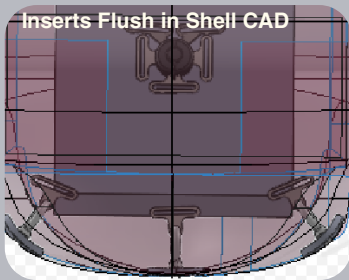
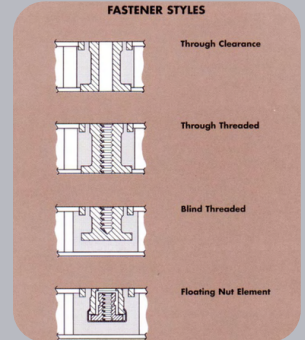
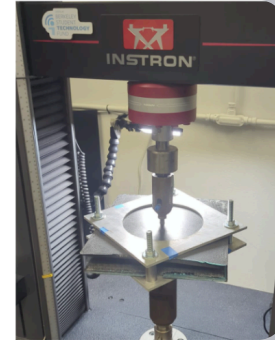
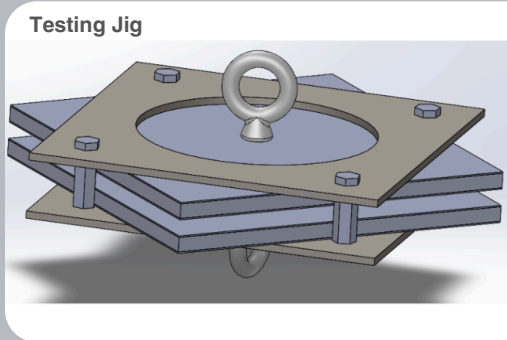
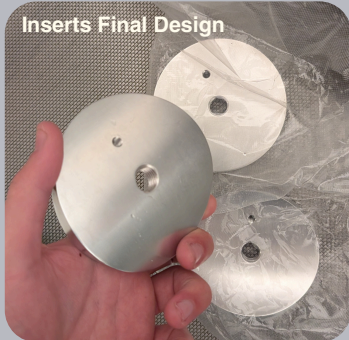
Performance & Validation

- Achieved 0.72 kg max safe payload in **pick-and-place tests**, a 1.44x safety factor over the 0.5 kg design target
- Full setup from disassembled state averaged ~ 10 minutes including firmware; mechanical assembly in 2 min 20 s
- Toolless cup/tube replacement **across multiple operators** averaged ~ 30 s

Limitations & Future Improvements

- Long-run endurance testing still pending to confirm lifetime targets
- PCB assembly cost is the main bottleneck; single-sided, hand-solderable layout can cut cost (but larger design)
- Cup and enclosure geometry tuned to one object size class; other sizes would need separate cup variants
- Enclosure and mount assume a specific flange style and cable routing, so some arms would need design revisions

PROJECT PORTFOLIO



CALSOL: LAP & ANTI-SUBMARINE BELTS

Overview

As Driver Safety Lead for CalSol's GenXI solar vehicle, I designed the lap and sub-belt **anchorage system** end to end: bonded metal inserts that transfer concentrated harness loads into a thin carbon-fiber monocoque without loading the shell directly. Validated the design with analysis and a **purpose-built quasi-static pull-out test**.

- Part of the five-point harness system that became the first mechanical subsystem to pass 2026 race cycle scrutineering
- Result: cleared analytical load cases; pull-out testing across four samples corresponding to 1.6 FOS

Design Architecture & Choices

- **Bonded metal inserts** shaped to match the shell curvature, sitting flush within the laminate to spread load
- Insert **threaded through**, option to add a nut if necessary while keeping exterior aerodynamic otherwise
- **Eyebolt** connects to a clip holding the seatbelts, easy instalment
- Safety wire added as secondary retention against vibration loosening

Performance & Validation

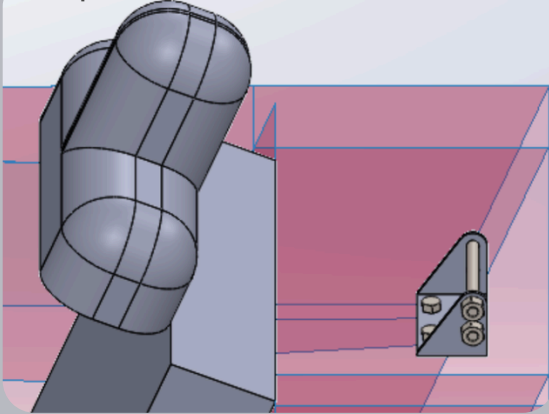
- Analyzed thread stripping and laminate debonding as the primary failure modes
- Built a quasi-static pull-out test jig and tested four samples, measuring an average failure at ~70% of the regulated load
 - Treated test as a deliberately conservative proxy for dynamic loading since crash-sled equipment wasn't available
 - Test setup revised after results indicated slippage (see image bottom right)
 - Static pull out underestimates dynamic capacity; combined with published data on bonded CFRP behavior under crash loading, the result supported a credible case for meeting the real impact requirement with 1.6 FOS

Limitations & Future Improvements

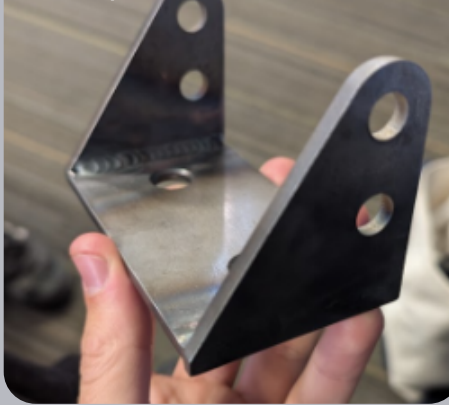
- Design is chassis-specific to GenXI car; a more parametric approach would ease adaptation to future car generations
- Used quasi-static pull test and literature to validate safety-critical system, actual dynamic pull test would be best
- Insert geometry already requires multi-axis machining; a clever design could further reduce mass

PROJECT PORTFOLIO

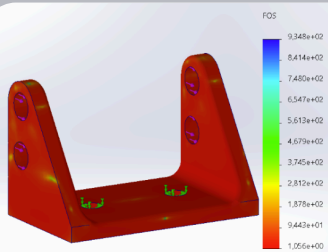
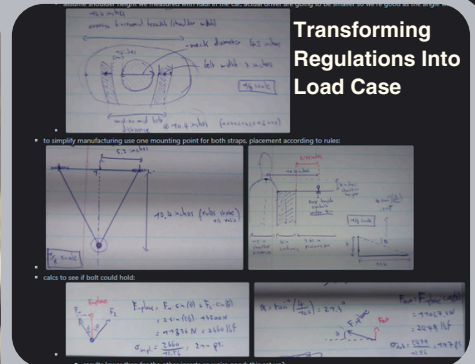
Concept CAD



Test Sample



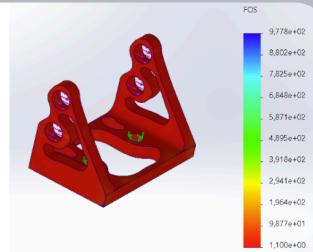
Transforming Regulations Into Load Case



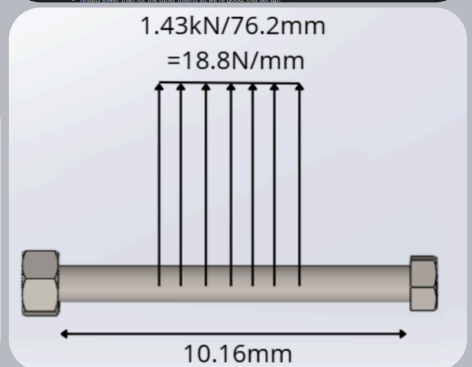
FOS Simulation pre-optimization



Proposed mesh



FOS Simulation after optimization



CALSOL: SHOULDER BELTS

Overview

As Driver Safety Lead for CalSol's GenXI solar vehicle, I owned the shoulder-belt anchorage end to end, translating race regulations into load cases before designing, analyzing and weight-optimizing a **welded steel mount bolted into the chassis**.

- Part of the five-point harness system that became the first mechanical subsystem to pass 2026 race cycle scrutineering
- Result: governing load case cleared with margin, ~40% weight reduction via topology optimization

Design Architecture

- Two bolts attaching anchorage to the chassis
- Three part anchorage from single waterjet-cut plate, welded assembly
- Transverse wrapping bolts through the anchorage to hold belts

Design Choices

- Kept cost low by using off-the-shelf hardware and making in house manufacturable
- Straps wrapped independently, designing out single-point failure
- Selected annealed 4130 steel for weldability, strength, and ductile failure behavior in a safety-critical weld
- Ran topology optimization for weight minimization, constrained to stay uniform-thickness, waterjet-friendly and weldable

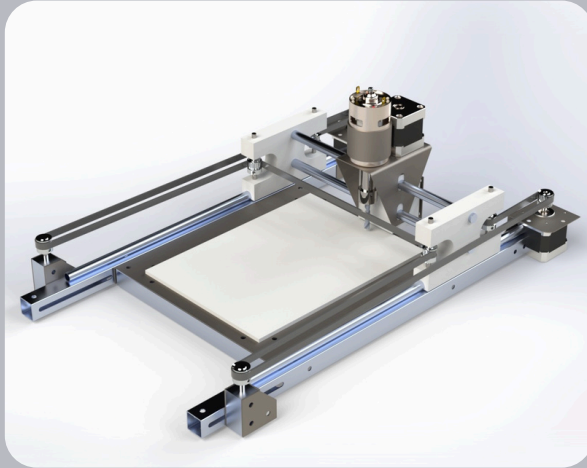
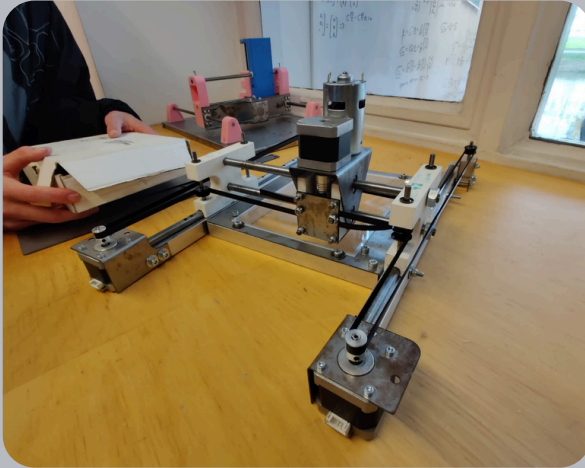
Performance & Validation

- Converted race regulations into explicit load cases for the anchorage
- Modeled wrapping bolts as fixed-fixed beams under distributed load; bolt bending = governing failure mode
- Validated the anchorage geometry with FEA before applying topology optimization

Limitations & Future Improvements

- Simulation cleared the load cases, physical pull test needed to confirm modeled behavior before race use
- A full-harness dynamic sled test is custom for a safety-critical system

PROJECT PORTFOLIO



DESIGNING AND BUILDING A CNC MILL

Overview

Designed and built a compact **PMMA-only** CNC mill with an **H-bot gantry**, GRBL-based motion control, and an aluminum/sheet-metal frame. Validated performance through **homing checks**, **test cuts**, and **resonance tuning** to improve repeatability and cut quality.

- Result: 10% smaller footprint, reduced part count relative to a faculty benchmark design, top 5% in competition

Design Architecture

- **Gantry:** H-bot, single-belt layout; fewer motion parts, stationary motors reduce moving mass, travel speed increased
- **Z-axis:** lead screw plus dual guide bars for vertical stiffness and positioning accuracy
- **Spindle:** RS-550 12V drill motor at ~18,000 RPM
- **Motors:** NEMA 17 steppers (1.8° step, 200 steps/rev, 1.7 A/phase max, ~0.4–0.6 N·m holding torque)
- **Electronics:** Arduino Uno R3, CNC Shield v3 board, A4988 drivers, GRBL + Universal G-code Sender, 12V/2A supply

Design Choices

- Chose a simple, low-cost electronics stack (~\$120) over a more complex motion controller
- Integrated a **modular belt pre-tensioning** system by repositioning structural mounts in slots on the aluminum extrusion base
- Accuracy = system-level property: motion tuning, belt tension, frame stiffness, workholding, and vibration addressed together
- Added mass to the frame to shift resonant frequencies away from the operating range, verified via oscilloscope vibration testing across spindle RPM

Control & Software

- GRBL firmware on Arduino, controlled via Universal G-code Sender (UGS)
- Tuned motor steps/mm, feed rate, acceleration, spindle speed, and chip load for PMMA

Performance & Validation

- Achieved ~±0.25 mm general XY accuracy; improved to ~±0.13 mm on critical features by reducing feed speed ~40%
- Process parameters: 2 mm depth of cut, 500–800 mm/min feed rate, 0.014–0.022 mm/tooth chip load
- Tuned speed by ramping up until motor/gantry skipping occurred, then backing off to a stable margin
- Quantified accuracy through repeated homing and offset measurement
- Verified X/Y consistency with a largest-square test

Limitations & Future Improvements

- No limit switches, needed for safety and overtravel protection
- Limited Z travel range and material compatibility (PMMA-focused, not general-purpose)
- H-bot architecture requires ongoing attention to gantry racking and rail lubrication
- Accuracy is strong for PMMA sheet work but below the range of a rigid general-purpose mill



THANK YOU

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